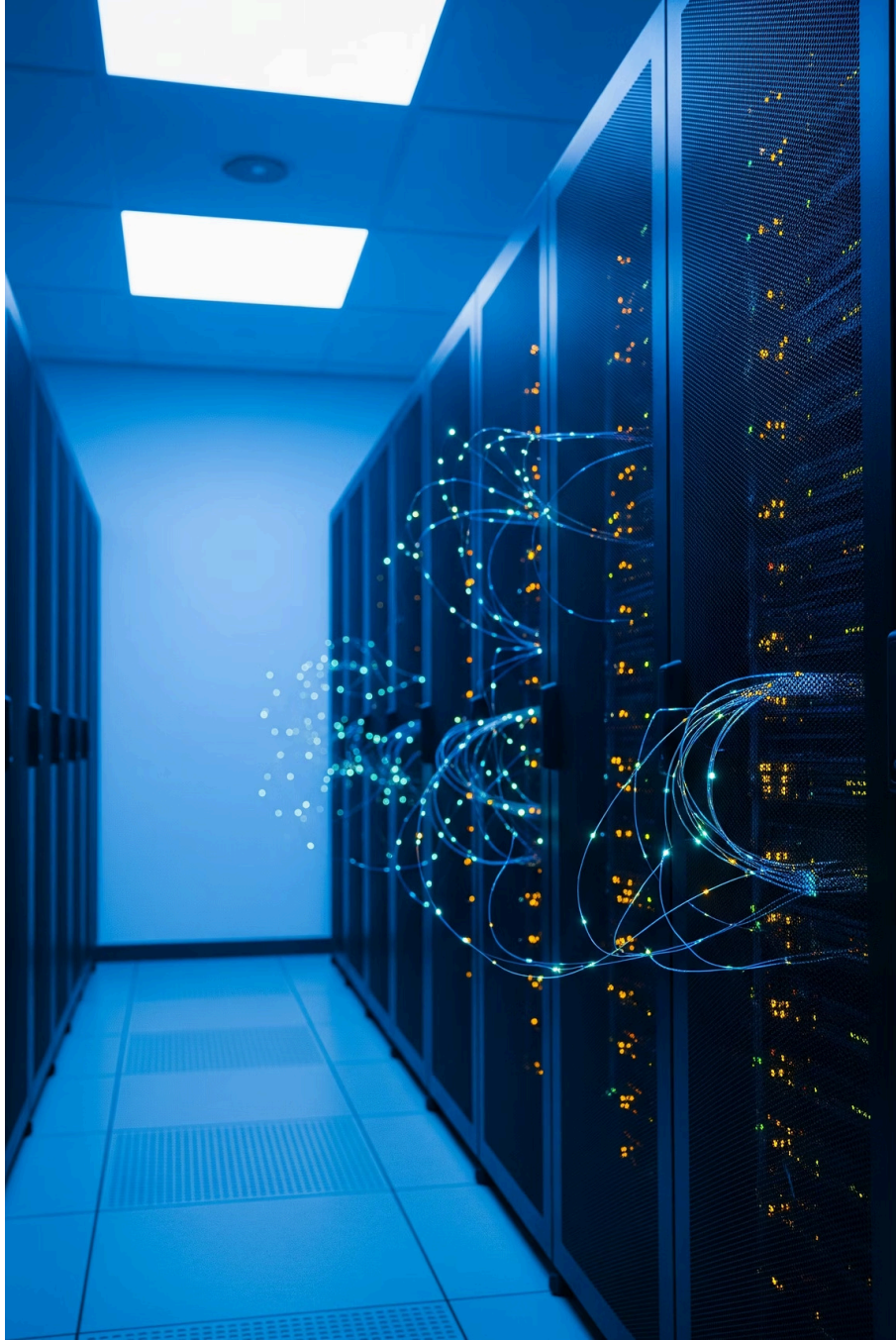
Ads Genie: Implementation Plan

Prepared for: Render Analytics
Prepared by: Chris Morancie — Fractional CTO / AI Architect
Date: March 2026

DRAFT — FOR DISCUSSION



Overview

This plan translates the architecture recommendations from the companion Architecture Review into a buildable roadmap. The system is built in **three layers**, delivered across **six two-week sprints**.

260

Total Dev Hours

Estimated development effort across all phases

3

Layers

MCP Server → Orchestration → Slack Control Plane

Build Summary

Six build phases spanning the full system, from API integration to market validation.

Phase	What Gets Built	Dev Hours	Kal's Role
Layer 1: Custom MCP Server	Google Ads API wrapper (OAuth, GAQL, caching, rate limiting). 5 core MCP tools: health_check, analyze_budget_waste, diagnose_roas_drop, search_terms_audit, benchmark_account. Each tool encodes Render's methodology, not raw queries.	60 hrs.	Define thresholds, playbooks, waste definitions, and vertical benchmarks. This is the knowledge the tools encode.
Layer 2: Event-Driven Orchestration	EventBridge schedules (hourly health checks, Monday reports). Step Functions anomaly response workflow (detect → analyze → recommend → approve → execute → log). DynamoDB account registry with per-client autonomy config. Decision log table.	50 hrs.	Define severity thresholds, autonomy levels per action type, alert frequency limits, weekly report format.
Layer 3: Slack Control Plane	Slack bot (Bolt SDK, Socket Mode). Interactive alert messages with Approve/Modify/Dismiss buttons. Task token bridge (Slack approval → Step Functions continues). On-demand query support (@adsgenie in client channels). Per-client channel isolation.	40 hrs.	Test alert quality — are they actionable or noisy? Validate the full loop: alert → approve → action → confirmation.
Memory & Reporting	Decision memory (immutable audit trail per action). Client context memory (seasonal patterns, constraints, past outcomes). Automated weekly MCC report (Monday 8am). Per-client monthly reports.	40 hrs.	Define report format. Validate that memory context improves agent reasoning.
Campaign Builder & Write Actions	draft_campaign MCP tool (STAG methodology, vertical defaults). Write action tools (add negatives, pause/enable ad groups, adjust bids). All mutations gated by graduated autonomy, nothing executes without the right approval level.	40 hrs.	Review every draft campaign the system produces. Tune autonomy levels based on trust.
Expand & Validate	5 additional MCP tools (competitor analysis, landing page audit, keyword expansion, ad copy performance, pacing). 4 weeks of daily use validation. Cost model tracking.	30 hrs.	Daily use for 4 weeks. Track signal-to-noise ratio, time saved, accuracy. Go/no-go on market version.

Architecture Principles

Every decision in this plan maps to the AI Production Framework. Five core principles govern how the system is designed and operated.

Probabilistic reasoning inside deterministic infrastructure

The LLM picks tools and interprets results. Step Functions owns the workflow. MCP tools execute Render's logic exactly.

Routing is infrastructure, not AI

EventBridge routes events. Autonomy levels determine approval paths. The LLM does not decide its own permissions.

Every decision must be replayable

Full context logging; prompt, tool calls, inputs, outputs, human decision, timestamp. Immutable audit trail.

Identity and permissions are architectural

Non-human identity per Lambda. Least privilege. Secrets Manager for all credentials. No shared keys.

Human authority is designed, not assumed

HITL is a Step Functions wait state + task token, not a prompt instruction. Autonomy levels stored in DynamoDB, enforced by infrastructure.

Delivery Timeline

Six sprints across three architectural layers, each building on the last.

1

Milestone 1

MCP Server Core

Google Ads API wrapper, OAuth, GAQL, caching, rate limiting. First 5 MCP tools live.

2

Milestone 2

Orchestration Layer

EventBridge schedules, Step Functions workflow, DynamoDB registry. Anomaly detection pipeline.

3

Milestone 3

Slack Control Plane

Bolt SDK bot, interactive alerts, task token bridge, per-client channel isolation.

4

Milestone 4

Memory & Reporting

Decision memory, client context memory, weekly MCC reports, monthly per-client reports.

5

Milestone 5

Campaign Builder

draft_campaign tool, write actions (negatives, bids, ad groups), graduated autonomy gates.

6

Milestone 6

Expand & Validate

5 additional tools, 4 weeks daily use, cost tracking, go/no-go decision.

Technology Stack

A fully cloud-native, Python-first stack built on AWS infrastructure with Claude as the reasoning engine.

Component	Technology
Custom MCP Server	Python (MCP SDK or FastAPI)
Google Ads API	Python client library (v18+)
LLM	Claude API (Sonnet for standard, Opus for complex diagnosis)
Orchestration	AWS Step Functions
Scheduling / Events	Amazon EventBridge
Compute	AWS Lambda

Component	Technology
Database	Amazon DynamoDB
Slack Bot	Slack Bolt SDK (Python)
Audit Logs	S3 with Object Lock
Secrets	AWS Secrets Manager
Observability	CloudWatch + X-Ray
IaC	AWS CDK (Python)

Security Architecture

Security is not bolted on, it is designed into every layer of the system from day one.

Three-Point AI Firewall

Validate **inbound** (Slack messages), **outbound** (LLM-proposed actions against autonomy config), and **external** (Google Ads API responses) before any processing occurs.

Separate IAM Roles

Distinct roles for the monitoring Lambda, action-execution Lambda, and Slack handler. Least-privilege enforced at the infrastructure level, not by convention.

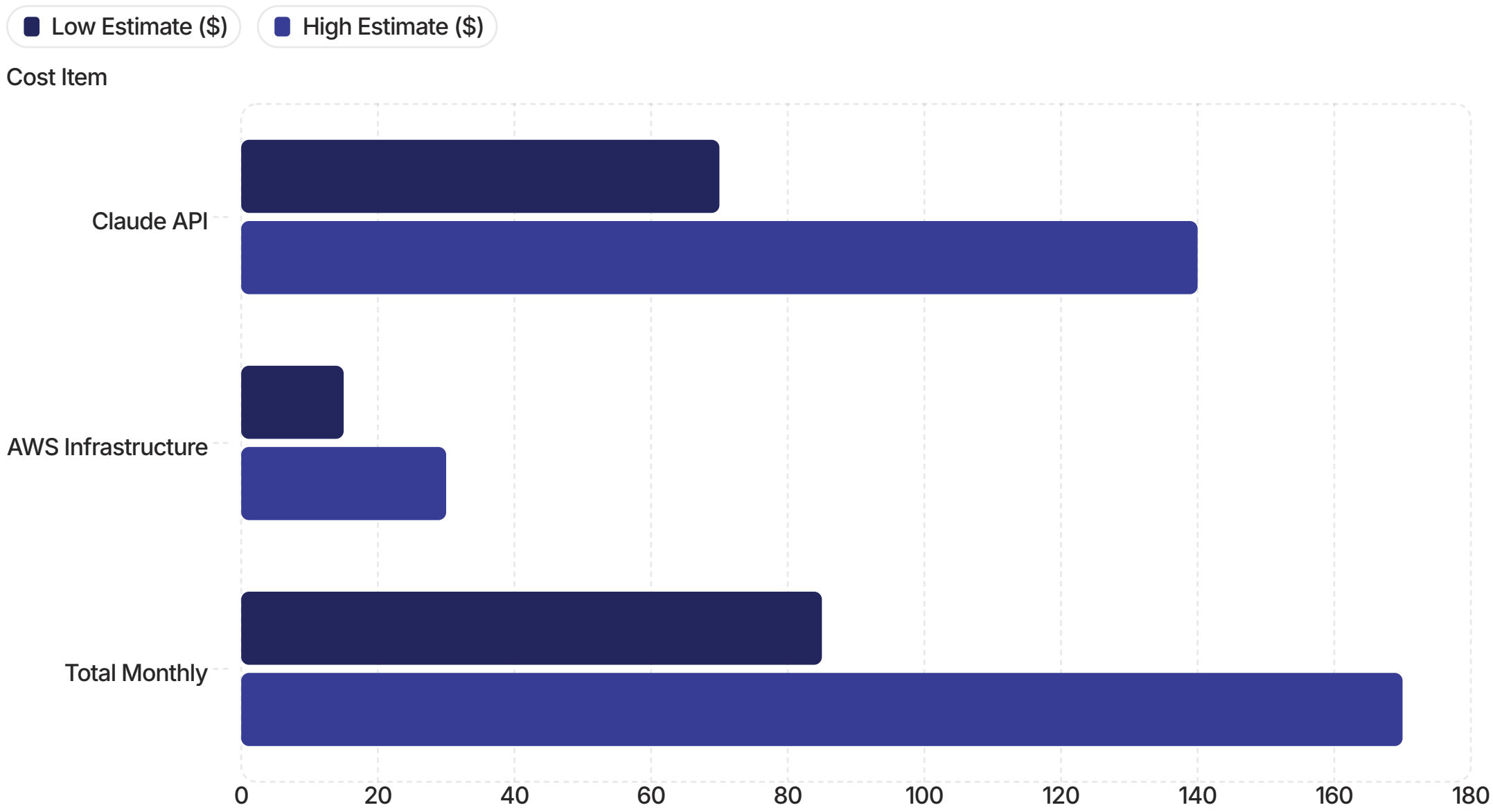
Zero Hardcoded Secrets

All credentials; Google OAuth, Slack tokens, Claude API key are stored in AWS Secrets Manager. No secrets in code, environment variables, or configuration files.

Cost Estimate — Internal Use

Lean infrastructure costs make this profitable from the very first client at Solo tier pricing.

Item	Monthly Cost
Claude API (Sonnet + Opus)	~$70–140
AWS (Lambda, Step Functions, DynamoDB, EventBridge, S3)	~$15–30
Total	**~$85–170/month**



Key Risks & Mitigations

Five identified risks with concrete mitigation strategies built into the plan from the start.

Risk	Mitigation
Google Ads Developer Token delay	Apply immediately. Basic access is fast. Standard access takes longer. Early phases are read-only anyway, so development is not blocked.
MCP tools don't match Kal's judgment	Golden dataset testing against 10 real accounts. Iterate until tool outputs match manual analysis before moving to production use.
Alert fatigue	Start conservative. Configurable thresholds, deduplication, quiet hours. Alert less, not more with a focus on quality over quantity from day one.
Claude API cost creep	Per-call tracking from day one. Model cascade (Sonnet first, Opus only for complex diagnosis). Monthly budget caps enforced at infrastructure level.
Scope creep to market too early	Internal milestone gate: 4 weeks daily use and greater than 80% alert accuracy required before any market discussion begins.

Success Criteria

Three milestone gates define what "done" looks like at each phase of the build.

MCP Server Live

MCP server running. Kal querying real accounts via Claude Desktop. Tool outputs matching manual analysis on golden dataset accounts.

Always-On Monitoring

Always-on monitoring live. Slack alerts with approval workflow functional. Weekly reports generating automatically every Monday at 8am.

Go/No-Go Decision

Campaign builder working. Write actions executing through approvals. 4 weeks of daily use data collected. Go/no-go decision on market version made with full data.

Built on the AI Production Framework

10 pillars. 3 layers. 140+ principles for production-grade AI systems.

This implementation plan is a companion document to the Architecture Review. Every build decision, every infrastructure choice, and every security control in this plan traces back to a specific principle in the AI Production Framework, ensuring that Ads Genie is not just functional, but production-grade from day one.

Deterministic Infrastructure

Step Functions, EventBridge, and DynamoDB own all workflow logic

Human-in-the-Loop by Design

Approval gates are infrastructure wait states, not prompt instructions

Full Auditability

Every action logged with full context; immutable, replayable, reviewable

Graduated Autonomy

Trust is earned incrementally. Nothing executes without the right approval level

Prepared by Chris Morancie · March 2026